

Stratified EEG Connectivity Analysis

2026-05-31

load demographics & connectivity data

```
demo <- read_csv(file.path(base_dir, "Participants_LEMON.csv")) %>%
  rename(subject = `...1`) %>%
  mutate(
    subject = str_remove(subject, "^sub-"),
    subject = str_trim(subject),

    age_low = as.numeric(str_extract(Age, "^\\d+")),
    age_high = as.numeric(str_extract(Age, "\\d+$")),
    age_mid = (age_low + age_high) / 2,

    age_group_main = if_else(age_mid <= 60, "young", "old"),

    age_group_ablation = case_when(
      age_mid < 35 ~ "young",
      age_mid > 60 ~ "old",
      TRUE ~ NA_character_
    )
  )

## New names:
## Rows: 228 Columns: 3
## -- Column specification
## ----- Delimiter: "," chr
## (2): ...1, Age dbl (1): Gender_1=female_2=male
## i Use 'spec()' to retrieve the full column specification for this data. i
## Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## * ' -> '...1'
```

```
read_conn <- function(inv_method) {
  path <- file.path(
    base_dir,
    paste0("connectivity_", inv_method),
    "stats",
    "bootstrap_connectivity_summary.tsv"
  )

  read_tsv(path) %>%
    mutate(
      inverse_method = inv_method,
      subject = str_remove(subject, "^sub-"),

```

```

    subject = str_trim(subject)
  )
}

conn <- bind_rows(
  read_conn("dSPM"),
  read_conn("eLORETA")
)

## Rows: 864 Columns: 7
## -- Column specification -----
## Delimiter: "\t"
## chr (4): subject, task, band, method
## dbl (3): bootstrap_mean, bootstrap_std, bootstrap_snr
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.
## Rows: 252 Columns: 7
## -- Column specification -----
## Delimiter: "\t"
## chr (4): subject, task, band, method
## dbl (3): bootstrap_mean, bootstrap_std, bootstrap_snr
##
## i Use 'spec()' to retrieve the full column specification for this data.
## i Specify the column types or set 'show_col_types = FALSE' to quiet this message.

conn2 <- conn %>% left_join(demo, by = "subject")

conn2 %>% count(age_group_main)

## # A tibble: 2 x 2
##   age_group_main     n
##   <chr>          <int>
## 1 old             240
## 2 young           876

conn2 %>% count(age_group_ablation)

## # A tibble: 2 x 2
##   age_group_ablation     n
##   <chr>          <int>
## 1 old             240
## 2 young           876

demo %>%
  filter(
    age_mid >= 35,
    age_mid <= 60
  ) %>%
  select(subject, Age, age_mid)

```

```
## # A tibble: 5 x 3
##   subject Age   age_mid
##   <chr>   <chr>   <dbl>
## 1 032378 55-60    57.5
## 2 032391 35-40    37.5
## 3 032463 55-60    57.5
## 4 032482 55-60    57.5
## 5 032486 55-60    57.5
```

```
conn %>%
  semi_join(
    demo %>%
      filter(age_mid >= 35,
             age_mid <= 60),
    by = "subject"
  ) %>%
  count(subject)
```

```
## # A tibble: 0 x 2
## # i 2 variables: subject <chr>, n <int>
```

Stratified Wilcoxon age tests

```
age_tests_main_stratified <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
    !is.na(age_group_main)
  ) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_main == "young"),
    n_old = sum(age_group_main == "old"),
    mean_young = mean(bootstrap_mean[age_group_main == "young"], na.rm = TRUE),
    mean_old = mean(bootstrap_mean[age_group_main == "old"], na.rm = TRUE),
    p = ifelse(
      n_young > 0 & n_old > 0,
      wilcox.test(bootstrap_mean ~ age_group_main, exact = FALSE)$p.value,
      NA_real_
    ),
    .groups = "drop"
  ) %>%
  group_by(inverse_method, method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)

age_tests_main_stratified
```

```
## # A tibble: 24 x 10
##   inverse_method method   band task n_young n_old mean_young mean_old p
```

```
##      <chr>          <chr>      <chr> <chr>    <int> <int>      <dbl>      <dbl> <dbl>
## 1 eLORETA         imcoh       alpha Eyes~      7      3      0.0245     0.0183 0.0682
## 2 eLORETA         wpli2_de~ alpha Eyes~      8      3      0.129      0.110 0.262
## 3 eLORETA         wpli2_de~ alpha Eyes~      7      3      0.110      0.0831 0.254
## 4 eLORETA         wpli2_de~ beta  Eyes~      8      3      0.0435     0.0498 0.262
## 5 eLORETA         imcoh       alpha Eyes~      8      3      0.0279     0.0240 0.185
## 6 dSPM            wpli2_de~ alpha Eyes~     29      7      0.129      0.124 0.522
## 7 dSPM            wpli2_de~ alpha Eyes~     29      7      0.0991     0.0821 0.263
## 8 dSPM            wpli2_de~ beta  Eyes~     29      7      0.0481     0.0550 0.299
## 9 dSPM            wpli2_de~ beta  Eyes~     29      7      0.0451     0.0537 0.358
## 10 dSPM           wpli2_de~ theta Eyes~     29      7      0.0611     0.0552 0.603
## # i 14 more rows
## # i 1 more variable: p_fdr_stratified <dbl>
```

```
age_tests_ablation_stratified <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
    !is.na(age_group_ablation)
  ) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_ablation == "young"),
    n_old = sum(age_group_ablation == "old"),
    mean_young = mean(bootstrap_mean[age_group_ablation == "young"], na.rm = TRUE),
    mean_old = mean(bootstrap_mean[age_group_ablation == "old"], na.rm = TRUE),
    p = ifelse(
      n_young > 0 & n_old > 0,
      wilcox.test(bootstrap_mean ~ age_group_ablation, exact = FALSE)$p.value,
      NA_real_
    ),
    .groups = "drop"
  ) %>%
  group_by(inverse_method, method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)

age_tests_ablation_stratified
```

```
## # A tibble: 24 x 10
##   inverse_method method   band task n_young n_old mean_young mean_old p
##   <chr>          <chr>   <chr> <chr>   <int> <int>      <dbl>      <dbl> <dbl>
## 1 eLORETA         imcoh     alpha Eyes~      7      3      0.0245     0.0183 0.0682
## 2 eLORETA         wpli2_de~ alpha Eyes~      8      3      0.129      0.110 0.262
## 3 eLORETA         wpli2_de~ alpha Eyes~      7      3      0.110      0.0831 0.254
## 4 eLORETA         wpli2_de~ beta  Eyes~      8      3      0.0435     0.0498 0.262
## 5 eLORETA         imcoh     alpha Eyes~      8      3      0.0279     0.0240 0.185
## 6 dSPM            wpli2_de~ alpha Eyes~     29      7      0.129      0.124 0.522
## 7 dSPM            wpli2_de~ alpha Eyes~     29      7      0.0991     0.0821 0.263
## 8 dSPM            wpli2_de~ beta  Eyes~     29      7      0.0481     0.0550 0.299
## 9 dSPM            wpli2_de~ beta  Eyes~     29      7      0.0451     0.0537 0.358
## 10 dSPM           wpli2_de~ theta Eyes~     29      7      0.0611     0.0552 0.603
## # i 14 more rows
```

```
## # i 1 more variable: p_fdr_stratified <dbl>
```

Although the demographic file contained a small number of subjects between 35 and 60, none of those subjects were present in the connectivity summary data. Therefore, the main age split (young ≤ 60 vs old > 60) and the ablation split (young < 35 vs old > 60) resulted in the same effective connectivity sample and produced identical statistical results.

```
# ks tests
```

```
ks_age_main <- conn2 %>%
  filter(band %in% bands, method %in% conn_types, !is.na(age_group_main)) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_main == "young"),
    n_old = sum(age_group_main == "old"),
    ks_p = ifelse(
      n_young > 0 & n_old > 0,
      ks.test(
        bootstrap_mean[age_group_main == "young"],
        bootstrap_mean[age_group_main == "old"]
      )$p.value,
      NA_real_
    ),
    .groups = "drop") %>%
  group_by(inverse_method, method) %>%
  mutate(ks_p_fdr = p.adjust(ks_p, method = "BH")) %>%
  ungroup() %>%
  arrange(ks_p_fdr)

ks_age_main
```

```
## # A tibble: 24 x 8
##   inverse_method method      band task      n_young n_old  ks_p ks_p_fdr
##   <chr>          <chr>      <chr> <chr>      <int> <int> <dbl> <dbl>
## 1 eLORETA      wpli2_debiased alpha EyesClosed    8     3 0.303    0.6
## 2 eLORETA      wpli2_debiased alpha EyesOpen      7     3 0.25     0.6
## 3 eLORETA      wpli2_debiased beta  EyesClosed    8     3 0.121    0.6
## 4 eLORETA      wpli2_debiased beta  EyesOpen      7     3 0.4      0.6
## 5 eLORETA      wpli2_debiased theta EyesClosed    8     3 0.709    0.709
## 6 eLORETA      wpli2_debiased theta EyesOpen      7     3 0.7      0.709
## 7 dSPM         wpli2_debiased beta  EyesClosed   29     7 0.141    0.848
## 8 eLORETA      imcoh          alpha EyesClosed    8     3 0.303    0.85
## 9 eLORETA      imcoh          alpha EyesOpen      7     3 0.167    0.85
## 10 eLORETA     imcoh          beta  EyesClosed    8     3 0.709    0.85
## # i 14 more rows
```

```
ks_age_ablation <- conn2 %>%
  filter(band %in% bands, method %in% conn_types, !is.na(age_group_ablation)) %>%
  group_by(inverse_method, method, band, task) %>%
  summarise(
    n_young = sum(age_group_ablation == "young"),
```

```

n_old = sum(age_group_ablation == "old"),
ks_p = ifelse(
  n_young > 0 & n_old > 0,
  ks.test(
    bootstrap_mean[age_group_ablation == "young"],
    bootstrap_mean[age_group_ablation == "old"]
  )$p.value,
  NA_real_
),
.groups = "drop") %>%
group_by(inverse_method, method) %>%
mutate(ks_p_fdr = p.adjust(ks_p, method = "BH")) %>%
ungroup() %>%
arrange(ks_p_fdr)

ks_age_ablation

```

```

## # A tibble: 24 x 8
##   inverse_method method      band task      n_young n_old  ks_p ks_p_fdr
##   <chr>          <chr>      <chr> <chr>      <int> <int> <dbl> <dbl>
## 1 eLORETA      wpli2_debiased alpha EyesClosed    8     3 0.303    0.6
## 2 eLORETA      wpli2_debiased alpha EyesOpen     7     3 0.25     0.6
## 3 eLORETA      wpli2_debiased beta  EyesClosed    8     3 0.121    0.6
## 4 eLORETA      wpli2_debiased beta  EyesOpen     7     3 0.4      0.6
## 5 eLORETA      wpli2_debiased theta EyesClosed    8     3 0.709    0.709
## 6 eLORETA      wpli2_debiased theta EyesOpen     7     3 0.7      0.709
## 7 dSPM         wpli2_debiased beta  EyesClosed   29     7 0.141    0.848
## 8 eLORETA      imcoh        alpha EyesClosed    8     3 0.303    0.85
## 9 eLORETA      imcoh        alpha EyesOpen     7     3 0.167    0.85
## 10 eLORETA     imcoh        beta  EyesClosed    8     3 0.709    0.85
## # i 14 more rows

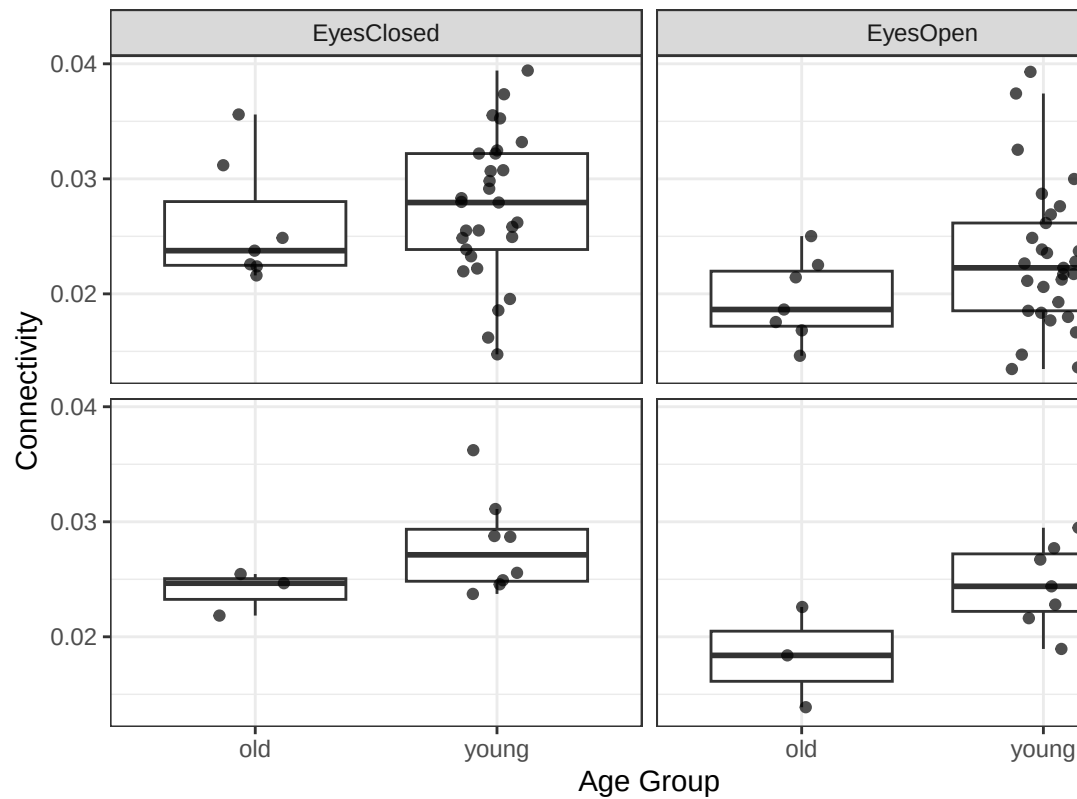
```

```

ggplot(
  conn2 %>%
    filter(band == "alpha", method == "imcoh", !is.na(age_group_ablation)),
  aes(x = age_group_ablation, y = bootstrap_mean)) +
  geom_boxplot(outlier.shape = NA) + geom_jitter(width = 0.15, alpha = 0.7) +
  facet_grid(inverse_method ~ task) + theme_bw() +
  labs(x = "Age Group", y = "Connectivity",
    title = "Alpha imcoh Connectivity: Young <35 vs Old >60"
  )

```

Alpha imcoh Connectivity: Young <35 vs Old >60



box plot with jitter points

ANOVA and ANCOVA

```
anova_data <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types,
    !is.na(age_group_ablation)
  )

anova_model <- lm(
  bootstrap_mean ~ age_group_ablation * band * task * inverse_method * method,
  data = anova_data
)

Anova(anova_model, type = 3)
```

Anova Table (Type III tests)

##

Response: bootstrap_mean

##

(Intercept)

age_group_ablation

band

task

	Sum Sq	Df	F value
(Intercept)	0.004730	1	16.0795
age_group_ablation	0.000012	1	0.0394
band	0.000746	2	1.2679
task	0.000147	1	0.4997

## inverse_method	0.000008	1	0.0288
## method	0.033946	1	115.3992
## age_group_ablation:band	0.000019	2	0.0327
## age_group_ablation:task	0.000013	1	0.0432
## band:task	0.000187	2	0.3183
## age_group_ablation:inverse_method	0.000010	1	0.0340
## band:inverse_method	0.000035	2	0.0601
## task:inverse_method	0.000001	1	0.0022
## age_group_ablation:method	0.000022	1	0.0760
## band:method	0.007708	2	13.1019
## task:method	0.002252	1	7.6571
## inverse_method:method	0.000176	1	0.5976
## age_group_ablation:band:task	0.000044	2	0.0752
## age_group_ablation:band:inverse_method	0.000011	2	0.0190
## age_group_ablation:task:inverse_method	0.000000	1	0.0001
## band:task:inverse_method	0.000006	2	0.0103
## age_group_ablation:band:method	0.000181	2	0.3068
## age_group_ablation:task:method	0.000158	1	0.5357
## band:task:method	0.002066	2	3.5120
## age_group_ablation:inverse_method:method	0.000125	1	0.4250
## band:inverse_method:method	0.000388	2	0.6593
## task:inverse_method:method	0.000121	1	0.4103
## age_group_ablation:band:task:inverse_method	0.000004	2	0.0063
## age_group_ablation:band:task:method	0.000294	2	0.5004
## age_group_ablation:band:inverse_method:method	0.000183	2	0.3108
## age_group_ablation:task:inverse_method:method	0.000012	1	0.0401
## band:task:inverse_method:method	0.000149	2	0.2531
## age_group_ablation:band:task:inverse_method:method	0.000076	2	0.1298
## Residuals	0.150020	510	
##	Pr(>F)		
## (Intercept)	6.982e-05	***	
## age_group_ablation	0.84264		
## band	0.28231		
## task	0.47997		
## inverse_method	0.86542		
## method	< 2.2e-16	***	
## age_group_ablation:band	0.96787		
## age_group_ablation:task	0.83551		
## band:task	0.72754		
## age_group_ablation:inverse_method	0.85368		
## band:inverse_method	0.94163		
## task:inverse_method	0.96293		
## age_group_ablation:method	0.78285		
## band:method	2.827e-06	***	
## task:method	0.00586	**	
## inverse_method:method	0.43986		
## age_group_ablation:band:task	0.92758		
## age_group_ablation:band:inverse_method	0.98115		
## age_group_ablation:task:inverse_method	0.99357		
## band:task:inverse_method	0.98975		
## age_group_ablation:band:method	0.73592		
## age_group_ablation:task:method	0.46455		
## band:task:method	0.03056	*	
## age_group_ablation:inverse_method:method	0.51472		


```
## band:inverse_method:method 0.51765
## task:inverse_method:method 0.52211
## age_group_ablation:band:task:inverse_method 0.99367
## age_group_ablation:band:task:method 0.60661
## age_group_ablation:band:inverse_method:method 0.73299
## age_group_ablation:task:inverse_method:method 0.84143
## band:task:inverse_method:method 0.77647
## age_group_ablation:band:task:inverse_method:method 0.87832
## Residuals
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
ancova_model <- lm(
  bootstrap_mean ~ age_mid * band * task * inverse_method * method,
  data = anova_data
)

Anova(ancova_model, type = 3)
```

```
## Anova Table (Type III tests)
##
## Response: bootstrap_mean
##
```

	Sum Sq	Df	F value	Pr(>F)	
## (Intercept)	0.005586	1	19.1085	1.497e-05	***
## age_mid	0.000004	1	0.0143	0.9047	
## band	0.001106	2	1.8926	0.1517	
## task	0.000025	1	0.0841	0.7720	
## inverse_method	0.000009	1	0.0314	0.8594	
## method	0.036819	1	125.9557	< 2.2e-16	***
## age_mid:band	0.000022	2	0.0372	0.9635	
## age_mid:task	0.000021	1	0.0720	0.7886	
## band:task	0.000005	2	0.0092	0.9909	
## age_mid:inverse_method	0.000014	1	0.0465	0.8293	
## band:inverse_method	0.000010	2	0.0166	0.9835	
## task:inverse_method	0.000001	1	0.0033	0.9544	
## age_mid:method	0.000000	1	0.0014	0.9700	
## band:method	0.009375	2	16.0360	1.761e-07	***
## task:method	0.000454	1	1.5521	0.2134	
## inverse_method:method	0.000074	1	0.2538	0.6147	
## age_mid:band:task	0.000057	2	0.0980	0.9066	
## age_mid:band:inverse_method	0.000014	2	0.0243	0.9760	
## age_mid:task:inverse_method	0.000000	1	0.0001	0.9939	
## band:task:inverse_method	0.000001	2	0.0020	0.9980	
## age_mid:band:method	0.000221	2	0.3779	0.6855	
## age_mid:task:method	0.000298	1	1.0189	0.3132	
## band:task:method	0.000161	2	0.2761	0.7588	
## age_mid:inverse_method:method	0.000186	1	0.6352	0.4258	
## band:inverse_method:method	0.000063	2	0.1085	0.8972	
## task:inverse_method:method	0.000015	1	0.0504	0.8224	
## age_mid:band:task:inverse_method	0.000004	2	0.0068	0.9932	
## age_mid:band:task:method	0.000454	2	0.7762	0.4607	
## age_mid:band:inverse_method:method	0.000237	2	0.4049	0.6673	
## age_mid:task:inverse_method:method	0.000016	1	0.0546	0.8153	
## band:task:inverse_method:method	0.000017	2	0.0297	0.9707	

```
## age_mid:band:task:inverse_method:method 0.000079 2 0.1347 0.8740
## Residuals 0.149082 510
## ---
## Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

dSPM vs eLORETA using paired testing Wilcoxon-Signed rank test

```
dspm_vs_eloreta <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types
  ) %>%
  select(subject, task, band, method, inverse_method, bootstrap_mean) %>%
  pivot_wider(
    names_from = inverse_method,
    values_from = bootstrap_mean
  ) %>%
  filter(!is.na(dSPM), !is.na(eLORETA)) %>%
  group_by(method, band, task) %>%
  summarise(
    mean_dSPM = mean(dSPM, na.rm = TRUE),
    mean_eLORETA = mean(eLORETA, na.rm = TRUE),
    p = wilcox.test(dSPM, eLORETA, paired = TRUE, exact = FALSE)$p.value,
    .groups = "drop"
  ) %>%
  group_by(method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)

dspm_vs_eloreta
```

```
## # A tibble: 12 x 7
##   method      band task      mean_dSPM mean_eLORETA      p p_fdr_stratified
##   <chr>      <chr> <chr>      <dbl>      <dbl> <dbl>      <dbl>
## 1 imcoh      theta EyesClos~ 0.0193      0.0197 0.0367      0.220
## 2 imcoh      beta  EyesClos~ 0.0101      0.00985 0.100       0.300
## 3 imcoh      alpha EyesClos~ 0.0271      0.0269 0.563       0.676
## 4 imcoh      alpha EyesOpen 0.0228      0.0227 0.476       0.676
## 5 imcoh      theta EyesOpen 0.0207      0.0209 0.359       0.676
## 6 wpli2_debiased theta EyesClos~ 0.0678      0.0693 0.120       0.718
## 7 imcoh      beta  EyesOpen 0.0103      0.0103 0.919       0.919
## 8 wpli2_debiased alpha EyesClos~ 0.124       0.124 0.965       1
## 9 wpli2_debiased alpha EyesOpen 0.102       0.102 1           1
## 10 wpli2_debiased beta  EyesClos~ 0.0457      0.0452 1           1
## 11 wpli2_debiased beta  EyesOpen 0.0463      0.0460 0.838       1
## 12 wpli2_debiased theta EyesOpen 0.0760      0.0766 0.359       1
```

WPLI-debiased vs IMCOH using paired testing Wilcoxon-Signed rank test

```
wpli_vs_imcoh <- conn2 %>%
  filter(
    band %in% bands,
    method %in% conn_types
  ) %>%
  select(subject, task, band, inverse_method, method, bootstrap_mean) %>%
  pivot_wider(
    names_from = method,
    values_from = bootstrap_mean
  ) %>%
  filter(!is.na(wpli2_debiased), !is.na(imcoh)) %>%
  group_by(inverse_method, band, task) %>%
  summarise(
    mean_wpli = mean(wpli2_debiased, na.rm = TRUE),
    mean_imcoh = mean(imcoh, na.rm = TRUE),
    p = wilcox.test(wpli2_debiased, imcoh, paired = TRUE, exact = FALSE)$p.value,
    .groups = "drop"
  ) %>%
  group_by(inverse_method) %>%
  mutate(p_fdr_stratified = p.adjust(p, method = "BH")) %>%
  ungroup() %>%
  arrange(p_fdr_stratified)
```

wpli_vs_imcoh

```
## # A tibble: 12 x 7
##   inverse_method band  task      mean_wpli mean_imcoh      p p_fdr_stratified
##   <chr>          <chr> <chr>      <dbl>      <dbl>    <dbl>      <dbl>
## 1 dSPM          alpha EyesClosed  0.128      0.0271  1.75e-7    0.000000175
## 2 dSPM          alpha EyesOpen   0.0958     0.0224  1.75e-7    0.000000175
## 3 dSPM          beta  EyesClosed  0.0494     0.0111  1.75e-7    0.000000175
## 4 dSPM          beta  EyesOpen   0.0467     0.0106  1.75e-7    0.000000175
## 5 dSPM          theta EyesClosed  0.0600     0.0174  1.75e-7    0.000000175
## 6 dSPM          theta EyesOpen   0.0640     0.0184  1.75e-7    0.000000175
## 7 eLORETA       alpha EyesClosed  0.124      0.0269  3.86e-3    0.00592
## 8 eLORETA       alpha EyesOpen   0.102      0.0227  5.92e-3    0.00592
## 9 eLORETA       beta  EyesClosed  0.0452     0.00985  3.86e-3    0.00592
## 10 eLORETA      beta  EyesOpen   0.0460     0.0103  5.92e-3    0.00592
## 11 eLORETA      theta EyesClosed  0.0693     0.0197  3.86e-3    0.00592
## 12 eLORETA      theta EyesOpen   0.0766     0.0209  5.92e-3    0.00592
```